NTNU

Lecture TIØ 4145 Corporate Finance

13.10.2025

JON RAGNAR VIGGEN

CFO AT ALVI GROUP AS

JON.RAGNAR.VIGGEN@ALVIGROUP.NO

Agenda 12:15 – 15:00

1) Introduction
2) Chapter 18: Capital Budgeting and Valuation with Leverage
3) Chapter 19: Valuation and Financial Modeling
4) Case study: Company valuation
5) Mandatory exercise

Introduction

Financial Advisory

DIGITAL FINANCE

Real-time key figures, better control, management and effectiveness through automation of manual processes and financial modelling.

VALUATION

Independent valuations based on recognized methodology in connection with private placements, share sale, acquisitions, demerger/merger or similar.

FINANCIAL DUE DILIGENCE

Financial analysis and review of the company, to ensure a better decision support in a sale and/or acquisition process.

INVESTMENT ANALYSIS

Profitability assessment of large, strategic investments, scenario and sensitivity analysis.

FINANCIAL MODELLING

Complete decision-making tool with key figure analysis and valuation, built on advanced financial modelling.

M&A

PARTNER SEARCH

National and international search for relevant partner candidates for acquisitions, sales and strategic alliances.

ACQUISITIONS AND SALES

Facilitating and support throughout the process – from analysis and partner search, through negotiations to implementation and closing.

MERGER AND DEMERGER

Consulting and support in the implementation of mergers, demergers, MBOs, LBOs and other transaction models.

PRIVATE PLACEMENTS

Planning and implementation of private placements directed at strategic and industrial investors.

TRANSACTION SUPPORT

Consulting and support in specified parts of an acquisition or sale process. For example, valuation, indicative bids, negotiation support or financial due diligence.

2014-2019: Buy-side advisory for Lindbak Gruppen

Familiebedriften starter året med nytt oppkjøp

Trondheimsselskapet Lindbak AS overtar konkurrent i Drammen. Samtidig overtar neste generasjon ledelsen i selskapet.

Opprykk: Jan Petter Lindbak overtok ved nyttår som administrerende direktør i Lindbak AS, en av to hovedvirksomheter i Lindbak Gruppen i Trondheim. FOTO: JENS PETTER SØRAA

Series of acquisitions

2019

2017 MODERNE ROM

2016 Finnmark Kontorservice as

2014 Pitney Bowes

2010 UniCom Kontorsenter

2007 HALL TORGERSEN AS

2016: Sell-side advisory for Austmann Bryggeri AS

Trondheims-bryggeriet Austmann måtte få inn en partner for å kunne fortsette veksten.

Austmann Bryggeri får Hansa Borg som medeier

Oppkjøp: – Å ha et bryggeri i Midt-Norge er en spennende mulighet, sier Lars Andreas Midtgaard i Hansa Borg. Fra venstre: Anders Cooper i Austmann, Lars Andreas Midtgaard i Hansa Borg, Thomas Sjue og Vinko Lien Sindelar i Austmann. Foto: NILS H. TOLDNES

2020: Buy-side and sell-side advisory for Austmann Bryggeri AS

En av stifterne av E.C. Dahls bryggeri blir ny eier i Austmann

Nye eiere i Austmann gjør at bryggeriet seiler opp og kan bli mer enn et småeksotisk mikrobryggeri.

Nye medeiere: Austmann bryggeri har fått tre nye eksterne eiere, i tillegg til gründerne. Fra venstre Ole Petter Bjørseth, Jon-Ivar Holmslet, Thomas Sjue, Vinko Sindelar, Anne Strøm Nakstad og Anders Cooper. Bjørseth og Strøm Nakstad er nye eiere, sammen med Bjørn M. Wiggen.
FOTO: KIM NYGÅRD

Foto: Nils Heldal

Slik sikret Austmann kapital til å vokse under Corona-krisen

23. april kom nyheten om at det trønderske bryggeriet Austmann og giganten Hansa Borg skulle skille lag – midt under Corona-krisen. Knappe to dager senere er selskapet på ny i Adresseavisen, og denne gangn handler saken om nye eiere og vekstambisjoner. I løpet av en knapp måned har Austmann gjennomført en vellykket skillsmisse og rukket å gifte seg på ny.

Chapter 18: Capital Budgeting and Valuation with Leverage

Chapter 18: Capital Budgeting and Valuation with Leverage

Introduction

- Chapter 18 consider 3 methods for including the effect of leverage in the valuation of a company or project
  - WACC – weighted average cost of capital
  - APV – adjusted present value
  - FTE – flow to equity
- Other topics in this chapter
  - Project-based costs of capital
  - APV with other leverage policies
  - Other effects of financing
  - Advanced topics in capital budgeting

Chapter 18: Capital Budgeting and Valuation with Leverage

WACC – Weighted Average Cost of Capital

The WACC method: A stream of free cash flows discounted by the weighted average cost of capital, r_{wacc} or simply WACC, will obtain the levered value (Enterprise Value) of a company or project.

- Common assumptions
  - Constant debt to equity ratio
  - Corporate tax is the only imperfection
- Why use WACC?
  - A company obtain a tax shield by having interest bearing debt
  - The r_{wacc} includes the effect of tax in the valuation
  - If the company is fully financed with equity (zero debt) the r_{wacc} equals r_e
- The levered value (V^L) is given by the infinite free cash flows discounted by r_{wacc}

Chapter 18: Capital Budgeting and Valuation with Leverage

WACC – Weighted Average Cost of Capital

Example 1: Value a project with WACC

- I want to find the value of a suggested project
- Expected free cash flow:

Nordic Experience AS	**1**	**2**	**3**	**4**	**5**
Sales	0	5	5	5	5
Cost of goods	0	-2	-2	-2	-2
Gross profit	**0**	**3**	**3**	**3**	**3**
Operating expenses	-0,5	-1	-1	-1	-1
Depreciation	0	-0,3	-0,3	-0,3	-0,3
EBIT	**-0,5**	**1,7**	**1,7**	**1,7**	**1,7**
Tax (28%)	0,14	-0,48	-0,48	-0,48	-0,48
Unlevered net income	**-0,36**	**1,22**	**1,22**	**1,22**	**1,22**
Free cash flow estimation					
Depreciation	0	0,3	0,3	0,3	0,3
Capital expenditures	0	0	0	0	0
Increase in net working capital	0	0	0	0	0
Free cash flow	**-0,36**	**1,52**	**1,52**	**1,52**	**1,52**

Chapter 18: Capital Budgeting and Valuation with Leverage

WACC – Weighted Average Cost of Capital

- Finding the WACC:

Balance Sheet Nordic Experience AS

Numbers in MNOK

Assets		Liabilities and equity	
Cash	1	Debt	11
Assets	20	Equity	10
Total assets	21	Total liabilities and equity	21

- Remember to use the net debt (net debt = debt – cash)
- The WACC is computed by

$$r_{wacc} = [E/(E+D)] \cdot r_e + [D/(E+D)] \cdot r_d \cdot (1-\tau)$$

The levered value is given by discounting the free cash flow estimates:

$$V^L = \sum_{i=0}^{n} \frac{FCF_i}{(1+rwacc)^i}$$

Chapter 18: Capital Budgeting and Valuation with Leverage

WACC – Weighted Average Cost of Capital

Assumptions and calculations

- τ = corporate tax level = 27 %
- r_e = equity cost of capital = 10 %
- r_d = debt cost of capital = 5 %
- Initial investment = 3 MNOK

- Calculating the WACC
  - r_{wacc} = 10/20 · 10% + 10/20 · 5 % · (1-0,27) = 6,8 %

- Calculating the levered value V^L without the initial investment:
  - $V^L = \sum_{i=0}^{n} \frac{FCF_i}{(1+r_{wacc})^i}$ = 4,82 MNOK
- The net present value including the initial investment is 1,82 MNOK

Chapter 18: Capital Budgeting and Valuation with Leverage

WACC – Weighted Average Cost of Capital

Example 2: Value an acquisition with WACC

- Instead of investing in the project let us consider to buy another company
- Let us assume that I know the acquisition's effect on the free cash flow (FCF) and I also have an expected growth rate (g) for the business
- The acquisition can be valued as a growing perpetuity using the same WACC calculated earlier

- $V^L = \frac{FCF}{r_{wacc} - g}$

Chapter 18: Capital Budgeting and Valuation with Leverage

WACC – Weighted Average Cost of Capital

Assumptions and calculations

- FCF = 1 million
- g = 1 %
- Payment = 15 million

- Calculating the levered value V^L

$$V^L = \frac{FCF}{r_{wacc} - g} = \frac{1\ MNOK}{6{,}8\% - 1\%} = 17{,}24 \text{ MNOK}$$

- The net present value adjusted for the initial investment is 2,24 MNOK

Chapter 18: Capital Budgeting and Valuation with Leverage

WACC – Weighted Average Cost of Capital

Example 3: Adjusting for a constant debt to equity ratio

- Completing the project in Example 1 will add assets to my company with an initial market value of V_0^L = 4,82 MNOK
- Taking the initial investment of 3 MNOK into account the net present value of the project is 1,82 MNOK

Excercise: How can I obtain a constant debt to equity ratio in my example? Complete the calculations at home!

Hint: You can verify your debt capacity calculations by looking at example 4.

Chapter 18: Capital Budgeting and Valuation with Leverage

WACC – Weighted Average Cost of Capital

- The WACC method summarized:
  1. Determine the free cash flow of the investment
  2. Compute the Weighted Average Cost of Capital (r_{wacc} or simply WACC)
  3. Compute the value of the investment by discounting the cash flows with WACC

- The constant debt to equity ratio can be obtained by adjusting the debt and equity according to the free cash flows

- Using the WACC method is simple and straightforward, and is the most commonly method used for capital budgeting purposes

Chapter 18: Capital Budgeting and Valuation with Leverage

APV – Adjusted Present Value

The APV method: A valuation method where we determine the levered value by first calculating the unlevered value and then adding the value of the interest tax shield.

- V^L = APV = V^U + PV (interest tax shield)
- The APV method is more complicated than the WACC method because we have to compute both the V^U and the present value of the interest tax shield
- In addition, implementing the APV approach with a constant debt to equity ratio forces us to solve the project's debt and value simultaneously
- APV is preferable when the debt to equity ratio is not constant. Another positive aspect is that the method also shows the explicit value of the tax shield, which can be interesting information for managers and investors

Chapter 18: Capital Budgeting and Valuation with Leverage

APV – Adjusted Present Value

Example 4: Value a project with APV

- Let us apply the same assumptions as in example 1
- Expected free cash flow:

Nordic Experience AS	**1**	**2**	**3**	**4**	**5**
Sales	0	5	5	5	5
Cost of goods	0	-2	-2	-2	-2
Gross profit	**0**	**3**	**3**	**3**	**3**
Operating expenses	-0,5	-1	-1	-1	-1
Depreciation	0	-0,3	-0,3	-0,3	-0,3
EBIT	**-0,5**	**1,7**	**1,7**	**1,7**	**1,7**
Tax (28%)	0,14	-0,48	-0,48	-0,48	-0,48
Unlevered net income	**-0,36**	**1,22**	**1,22**	**1,22**	**1,22**
Free cash flow estimation					
Depreciation	0	0,3	0,3	0,3	0,3
Capital expenditures	0	0	0	0	0
Increase in net working capital	0	0	0	0	0
Free cash flow	**-0,36**	**1,52**	**1,52**	**1,52**	**1,52**

Chapter 18: Capital Budgeting and Valuation with Leverage

APV – Adjusted Present Value

Calculations

- Find the unlevered cost of capital (pretax WACC)

Balance Sheet Nordic Experience AS

Numbers in MNOK

Assets		Liabilities and equity	
Cash	1	Debt	11
Assets	20	Equity	10
Total assets	21	Total liabilities and equity	21

- $r_U = [E/(E+D)] \cdot r_e + [D/(E+D)] \cdot r_d = 10/20 \cdot 10\% + 10/20 \cdot 5\,\% = 7{,}5\,\%$
  - Note the difference from WACC: we do not adjust for the tax shield

- Compute the project's value without leverage:
  - $V^U = \sum_{i=0}^{n} \frac{FCF_i}{(1+r_U)^i} = 4{,}74$ MNOK

- Adjusting for the initial investment the value is 1,74 MNOK

Chapter 18: Capital Budgeting and Valuation with Leverage

APV – Adjusted Present Value

- Find the value of the tax shield by applying the debt capacity calculations from example 3 that you will do at home (see textbook for calculations of debt capacity):

	1	2	3	4	5
Debt capacity	2,59	2,01	1,38	0,71	
Interest paid		0,13	0,10	0,07	0,04
Tax shield		0,04	0,03	0,02	0,01

- Compute the present value of the tax shield
  - PV (tax shield) = $\sum_{i=0}^{n} \frac{Tax\ Shield_i}{(1+r_U)^i}$ = 0,08 MNOK

- The APV method:

  $V^L = V^U$ + PV (interest tax shield) = 1,74 MNOK + 0,08 MNOK = 1,82 MNOK

- As we can see (and as expected) the APV method give the same result as the WACC method

Chapter 18: Capital Budgeting and Valuation with Leverage

APV – Adjusted Present Value

Example 5: Value an acquisition with APV

- Apply the same assumptions as in example 2
- The APV method value the acquisition by computing the value without leverage and adding the present value of the tax shield
- $V^L = V^U + PV\ (\text{tax shield})$

- Value without leverage:
  - $V^U = \frac{FCF}{r_U - g} = \frac{1}{7{,}5\% - 1\%} = 15{,}38$

- To obtain the present value of the tax shield we have to keep in mind that we assume a constant debt to equity ratio. We know (from example 2) that the company will have to add 17,24 MNOK / 2 = 8,62 MNOK in debt to keep the ratio constant. Why is this? You should understand this before the exam!

Chapter 18: Capital Budgeting and Valuation with Leverage

APV – Adjusted Present Value

- Present value of tax shield with r_d = 5 %:
  - PV (tax shield) = $\frac{Tax\ Shield}{r_U - g} = \frac{8{,}62\ MNOK \times 5\ \% \times 27\ \%}{7{,}5\% - 1\%}$ = 1,86 MNOK

- The value obtained by the APV method is then calculated
  - $V^L = V^U$ + PV (tax shield) = 15,38 MNOK + 1,86 MNOK = 17,24 MNOK

- Adjusted for the initial investment we get a value of 2,24 MNOK

…which (of course) is the same result as in example 2

Chapter 18: Capital Budgeting and Valuation with Leverage

APV – Adjusted Present Value

- The APV method summarized:
  1. Determine the free cash flow of the investment
  2. Compute the value of the investment without leverage (V^U) by disconting the free cash flows with r_U
  3. Compute the present value of the tax shield
  4. Add the unlevered value V^U to the present value of the tax shield to determine the value of the investment with leverage V^L

Using the APV method is more complicated than the WACC method, but can be more appropriate if the company or project does not have a constant debt to equity ratio.

Chapter 18: Capital Budgeting and Valuation with Leverage

FTE – Flow to equity

The FTE method: This is a valuation method where we explicitly calculate the free cash flow available to the equity holders after taking into account all the payments to and from the debt holders.

- Instead of determining the free cash flow (FCF) we now determine the free cash flow to equity (FCFE)
  - The FCFE is the free cash flow that remains after adjusting for interest payments, debt issuance and debt repayments
  - Remember that we now use the equity cost of capital, r_e, as the discount rate
- There are two major differences from the free cash flow estimates we have used so far:
  - Interest expenses are deducted before taxes
  - The net borrowing activity is added as a separate line in the model

Chapter 18: Capital Budgeting and Valuation with Leverage

FTE – Flow to equity

Example 6: Flow to equity method

re =	10 %				
Nordic Experience AS	**1**	**2**	**3**	**4**	**5**
Sales	0	5	5	5	5
Cost of goods	0	-2	-2	-2	-2
Gross profit	**0**	**3**	**3**	**3**	**3**
Operating expenses	-0,5	-1	-1	-1	-1
Depreciation	0	-0,3	-0,3	-0,3	-0,3
EBIT	**-0,5**	**1,7**	**1,7**	**1,7**	**1,7**
Interest expense	0	-0,13	-0,10	-0,07	-0,04
Pretax income	**-0,50**	**1,57**	**1,60**	**1,63**	**1,66**
Tax (27%)	0,14	-0,44	-0,45	-0,46	-0,47
Net income	**-0,36**	**1,13**	**1,15**	**1,17**	**1,20**
Free cash flow to equity					
Depreciation	0	0,3	0,3	0,3	0,3
Capital expenditures	0	0	0	0	0
Increase in net working capital	0	0	0	0	0
Net borrowing	2,59	-0,59	-0,63	-0,67	-0,71
Free cash flow to equity	**2,23**	**0,85**	**0,83**	**0,81**	**0,79**
Present value	2,23	0,77	0,68	0,61	0,54
Net present value incl initial inv.	1,83				

Chapter 18: Capital Budgeting and Valuation with Leverage

FTE – Flow to equity

- The FTE method summarized:
  1. Determine the free cash flow to equity of the investment
  2. Determine the equity cost of capital, r_e
  3. Compute the contribution to equity value (E) by discounting the free cash flow to equity (FCFE) by the equity cost of capital (r_e)

- The FTE has the same disadvantage as the APV method as we also here need to compute the project's debt capacity to determinate the interest and net borrowing

- The FTE can offer an advantage if we are trying to compute the total value of the equity of a firm when the capital structure is complex

Chapter 18: Capital Budgeting and Valuation with Leverage

Conclusions

- WACC, APV and FTE all give the same results
- It is important to understand the WACC, APV and FTE methods to be able to choose the correct method for your specific case
- Different leverage policies, financial effects and periodical adjustments, might require alternative implementations of the methods

In real life, a lot of simplifications are made when it comes to valuation of companies. It is however crucial to know how to use the basic methods in this chapter so you understand what implications any form of simplification will have on the value of the company or the project subject to valuation!

Chapter 19: Valuation and Financial Modeling

Chapter 19: Valuation and Financial Modeling

Introduction

Valuation: a step-by-step guide

- Benchmark valuation using comparables
- The business plan and financial modelling
- Estimating the cost of capital
- Valuation
- Sensitivity analysis

Chapter 19: Valuation and Financial Modeling

Benchmark valuation using comparables

- One way to value a company is to compare multiples of similar companies
- Investors often use multiples as their main guideline when considering to buy a company
- The most common multiples are
  - P/E
  - EV/sales
  - EV/EBITDA
  - EBITDA/sales
  - EV/EBIT
- Alternative operational multiples: EV/customer, EV/hotel bed
- Different multiples are appropriate for different kind of businesses

When conducting a professional valuation, we always compare the multiples of our valuation with comparable companies in the market to see how our estimates fit with similar companies.

Chapter 19: Valuation and Financial Modeling

Benchmark valuation using comparables

- An example from the textbook (Ideko Corporation):

		Year 2005
	Income Statement ($ 000)	
1	**Sales**	75,000
2	Cost of Goods Sold	
3	Raw Materials	(16,000)
4	Direct Labor Costs	(18,000)
5	**Gross Profit**	41,000
6	Sales and Marketing	(11,250)
7	Administrative	(13,500)
8	**EBITDA**	16,250
9	Depreciation	(5,500)
10	**EBIT**	10,750
11	Interest Expense (net)	(75)
12	**Pretax Income**	10,675
13	Income Tax	(3,736)
14	**Net Income**	6,939

		Year 2005
	Balance Sheet ($ 000)	
	Assets	
1	Cash and Equivalents	12,664
2	Accounts Receivable	18,493
3	Inventories	6,165
4	**Total Current Assets**	37,322
5	Property, Plant, and Equipment	49,500
6	Goodwill	—
7	**Total Assets**	86,822
	Liabilities and Stockholders' Equity	
8	Accounts Payable	4,654
9	Debt	4,500
10	**Total Liabilities**	9,154
11	**Stockholders' Equity**	77,668
12	**Total Liabilities and Equity**	86,822

- Suggested price of the company: 150 million (Equity value)
- EV = Equity plus debt less cash in exess of working capital: 150 + 4,5 - 6,5 = 148 million (Enterprise value)

Chapter 19: Valuation and Financial Modeling

Benchmark valuation using comparables

- Comparing multiples of Ideko with similar companies in the market:

Ratio	Ideko (Proposed)	Oakley, Inc.	Luxottica Group	Nike, Inc.	Sporting Goods Industry
P/E		24.8×	28.0×	18.2×	20.3×
EV/Sales		2.0×	2.7×	1.5×	1.4×
EV/EBITDA		11.6×	14.4×	9.3×	11.4×
EBITDA/Sales		17.0%	18.5%	15.9%	12.1%

- By comparing multiples of similar companies we can establish an assumed range for the multiple of Ideko:

	Range	
Multiple	Low	High
P/E	18.2×	28.0×
EV/Sales	1.4×	2.7×
EV/EBITDA	9.3×	14.4×

Chapter 19: Valuation and Financial Modeling

Benchmark valuation using comparables

- Comparing multiples of Ideko with similar companies in the market:

Ratio	Ideko (Proposed)	Oakley, Inc.	Luxottica Group	Nike, Inc.	Sporting Goods Industry
P/E	21.6×	24.8×	28.0×	18.2×	20.3×
EV/Sales	2.0×	2.0×	2.7×	1.5×	1.4×
EV/EBITDA	9.1×	11.6×	14.4×	9.3×	11.4×
EBITDA/Sales	21.7%	17.0%	18.5%	15.9%	12.1%

- By comparing multiples of similar companies we can establish an assumed range for the multiple of Ideko:

	Range	
Multiple	Low	High
P/E	18.2×	28.0×
EV/Sales	1.4×	2.7×
EV/EBITDA	9.3×	14.4×

Chapter 19: Valuation and Financial Modeling

Benchmark valuation using comparables

	Range		Price (in $ million)	
Multiple	Low	High	Low	High
P/E	18.2×	28.0×	126.3	194.3
EV/Sales	1.4×	2.7×	107.0	204.5
EV/EBITDA	9.3×	14.4×	153.1	236.0

- Applying the P/E multiple on Ideko's earnings we get an interval for the implied value of the company. With earnings of 6.939 we get:
  - Low: P/E = 18,2 -> P = 126,3 M$
  - High: P/E = 28,0 -> P = 194,3 M$
- By applying Ideko's sales, EBITDA and earnings along with known multiples for similar companies, we can easily get an estimate for the Enterprise Value of Ideko
- Remember to adjust EV for NIBD to get the Equity value
- Such an approach will provide useful interval benchmark, but cannot be relied upon for a precise estimate of value

Chapter 19: Valuation and Financial Modeling

Benchmark valuation using comparables

- Example:

	Company A	Company B
Sales	1000	1000
Pretax margin	10%	8%
Pretax income	100	80
Income tax (30%)	30	24
Net income	**70**	**56**

Value of company A: 1400

Multiples for company A:
P/S = ?
P/E = ?

Multiples for company A applied to obtain the value of company B:

Value of company B with P/S = ?
Value of company B with P/E = ?

Chapter 19: Valuation and Financial Modeling

Benchmark valuation using comparables

- Example:

	Company A	Company B
Sales	1000	1000
Pretax margin	10%	8%
Pretax income	100	80
Income tax (30%)	30	24
Net income	**70**	**56**

Value of company A: 1400

Multiples for company A:

P/S =	?	1.4
P/E =	?	20

Multiples for company A applied to obtain the value of company B:

Value of company B with P/S = ?

Value of company B with P/E = ?

Chapter 19: Valuation and Financial Modeling

Benchmark valuation using comparables

- Example:

	Company A	Company B
Sales	1000	1000
Pretax margin	10%	8%
Pretax income	100	80
Income tax (30%)	30	24
Net income	**70**	**56**

Value of company A:	1400	
Multiples for company A:		
P/S =	?	1.4
P/E =	?	20

Multiples for company A applied to obtain the value of company B:

Value of company B with P/S =	?	1400
Value of company B with P/E =	?	1120

Chapter 19: Valuation and Financial Modeling

Benchmark valuation using comparables

- Example:

	Company A	Company B
Sales	1000	1000
Pretax margin	10%	8%
Pretax income	100	80
Income tax (30%)	30	24
Net income	**70**	**56**

Value of company A:	1400	
Multiples for company A:		
P/S =	?	1.4
P/E =	?	20

Multiples for company A applied to obtain the value of company B:

Value of company B with P/S =	?	1400
Value of company B with P/E =	?	1120

- Worth considering:
  - How comparable are the companies after all? Sector, group vs individual company, growth expectations, immaterial properties, business models etc.
  - Are we looking at the same value? Enterprise value vs Equity value.
  - Are the numbers comparable? Financial definitions, years, currency, tax situation etc.

Chapter 19: Valuation and Financial Modeling

Benchmark valuation using comparables

UTKAST

4. METODEVALG

Om valgte verdsettelsesmetoder

Normalresultatmetoden

- Normalresultat er basert på historisk regnskapsinformasjon, og er en tilnærming til fremtidige kontantstrømmer.
- Ved bruk av normalresultatmetoden etableres det et normalisert realjustert driftsresultat, som selskapet forventes å kunne levere gitt dagens kapasitet.
- Deretter diskonteres resultatet etter skatt med et **realjustert avkastningskrav** for å finne selskapsverdien. Dersom selskapet har lite omsettelige aksjer (for eksempel mindre selskap som ikke er børsnotert), trekkes en illikviditetsrabatt fra for å beregne faktisk selskapsverdi for det spesifikke selskapet som vurderes.
- Verdien av egenkapitalen estimeres deretter ved å justere selskapsverdien med likvide midler (+), ikke-driftsrelaterte eiendeler (+) og rentebærende/finansiell gjeld (-) og i noen tilfeller unormalt nivå på netto arbeidskapital (+/-)

SELSKAPSVERDI FØR ILLIKVIDITETSRABATT
- Illikviditetsrabatt
= SELSKAPSVERDI
+ Likvide midler og ikke driftsrelaterte aktiver
- Rentebærende/finansiell gjeld
+/- arbeidskapitaljustering
= VERDI AV EGENKAPITAL

Multippelbetraktninger

- Ved å studere sammenlignbare selskaper/bransjer kan man benytte observerbare nøkkeltall for å estimere verdien av Selskapet.
- Investorer benytter ofte multipler som en viktig benchmark når de vurderer hvorvidt de skal investere i et selskap eller ikke.
- Hvilke multipler som er relevante avhenger av type selskap og bransje. De mest vanlige multiplene er EV/EBITDA, P/E, EV/Sales, EV/EBIT og P/S.
- Som underlag for multippelberegningen danner man en peer group av sammenlignbare selskaper med kjente nøkkeltall. Deretter estimeres multiplene, og en snittbetraktning for peer group (gjennomsnitt eller median) kan benyttes som estimat for objektet som skal vurderes. Et eksempel på metodisk beregning er skissert nedenfor:

	EV/EBITDA		EV/EBIT		P/S	
Selskap #1	6,7		7,5		0,6	
Selskap #2	5,2		6,3		0,9	
Selskap #3	8,3		9,8		0,3	
Selskap #4	7,3		7,9		0,4	
Selskap #5	6,4		7,4		0,6	
Observasjon	Høy:	8,3	Høy:	9,8	Høy:	0,9
	Lav:	5,2	Lav:	6,3	Lav:	0,3
	Gj.snitt:	6,8	Gj.snitt:	7,8	Gj.snitt:	0,6
	Median:	6,7	Median:	7,5	Median:	0,6

MEMBER OF GLOBALSCOPE INTERNATIONAL M&A ADVISORS

IMPELLO Impello Management AS

Chapter 19: Valuation and Financial Modeling

The business plan and financial modelling

- Determining a realistic budget, including sales and operating costs, gives us the opportunity to obtain a more precise valuation
- The budget is often provided by the companies themselves, and has to be verified by the analyst
  - Beware that the company may have incentive for making the budget either expansive or defensive depending on the purpose of the valuation (!)
- A valuation is a subjective consideration depending highly on the motive:
  - Seller: from a seller's point of view the budgets are often expansive to obtain a highest possible value of the company
  - Buyer: from a buyer's point of view the budgets are often defensive to obtain a value as low as possible
  - Neutral: for tax purposes a neutral budget is used
- When entering negotiations you have to take such factors into account

Chapter 19: Valuation and Financial Modeling

The business plan and financial modelling

		2010	2011	2012	2013	Prognose 2014	2015	2016
						Prognose		
		2010	2011	2012	2013	2014	2015	2016
Sum driftsinntekter		5 144 386	5 295 622	6 053 394	6 356 064	6 673 867	7 007 560	7 357 938
Driftskostnader								
Varekostnader		(1 401 087)	(1 432 914)	(1 845 065)	(1 937 318)	(2 034 184)	(2 135 893)	(2 242 688)
Personalkostnader		(601 150)	(694 114)	(781 720)	(820 806)	(861 846)	(904 939)	(950 186)
Andre driftskostnader		(1 017 785)	(157 054)	(732 780)	(769 419)	(807 890)	(848 284)	(890 699)
EBITDA		**2 124 364**	**3 011 540**	**2 693 829**	**2 828 520**	**2 969 946**	**3 118 444**	**3 274 366**
Avskrivninger		(675 110)	(780 769)	(841 113)	(883 169)	(927 327)	(973 693)	(1 022 378)
EBIT		**1 449 254**	**2 230 771**	**1 852 716**	**1 945 352**	**2 042 619**	**2 144 750**	**2 251 988**
Skatt		(405 791)	(624 616)	(518 760)	(544 699)	(571 933)	(600 530)	(630 557)
Driftsresultat etter skatt		**1 043 463**	**1 606 155**	**1 333 956**	**1 400 653**	**1 470 686**	**1 544 220**	**1 621 431**
Av- og nedskrivninger		675 110	780 769	841 113	883 169	927 327	973 693	1 022 378
Endring i arbeidskapital	12,5 %				(37 834)	(39 725)	(41 712)	(43 797)
Investeringer								
Fri kontantstrøm					**2 245 988**	**2 358 288**	**2 476 202**	**2 600 012**

- We have a free cash flow estimate for 2013-2016 and can apply the WACC method to obtain the company's value.

Chapter 19: Valuation and Financial Modeling

Estimating the cost of capital

- To obtain the unlevered cost of capital we use the CAPM
- The cost of capital is given by:
  - $r = r_f + \beta(r_m - r_f) + \alpha = r_f + \beta \cdot RP + \alpha$ where $\beta = \frac{\rho_{R,Rm} \cdot \sigma_R}{\sigma_{Rm}}$
- CAPM and its components:
  - r_f: 3, 5 or 10 year governmental bonds are often used as the risk free rate
  - β: the beta is estimated from a peer group of similar companies
  - r_m: market risk which is empirically obtained
  - α: net adjustment for other relevant factors
  - $RP = r_m - r_f$ = risk premium
- The levered beta for the peer group companies are available through Reuters, Bloomberg, Yahoo Finance or similar
- Unlevering the beta:
  - $\beta_U = \frac{\beta_L}{(1+(1-t)*\left(\frac{D}{E}\right))}$

Chapter 19: Valuation and Financial Modeling

Estimating the cost of capital

- Example from the energy industry:

Selskap	Land	MCAP	D/E ratio	Skattesats	Beta levered	Beta unlevered
A2A SpA	Italia	16 330	1,21	31 %	0,93	0,51
Alpiq Holding AG	Sveits	20 963	0,90	18 %	0,61	0,35
Centralschweizerische Kraftwerke AG	Sveits	11 356	0,00	18 %	0,19	0,19
Centrica PLC	Storbritannia	189 562	0,81	23 %	0,48	0,30
E. ON AG	Tyskland	201 610	0,53	30 %	0,99	0,72
EDP-Energias de Portugal SA	Portugal	65 687	1,79	25 %	0,83	0,35
Electricite de France SA	Frankrike	320 931	1,33	33 %	0,91	0,48
Electricite de Strasbourg SA	Frankrike	5 456	0,03	33 %	0,52	0,51
Enbw Energie Baden Wuerttemberg AG	Tyskland	59 126	1,12	30 %	0,17	0,09
Endesa SA	Spania	151 763	0,35	30 %	0,86	0,69
Energiedienst Holding AG	Sveits	7 042	0,03	18 %	0,42	0,41
EVN AG	Østerrike	15 102	0,60	25 %	0,46	0,32
Fortum Oyj	Finland	110 951	0,73	25 %	0,70	0,45
Hafslund ASA	Norge	9 172	1,20	28 %	0,65	0,35
Hera SpA	Italia	15 517	1,26	31 %	0,65	0,35
Iberdrola SA	Spania	204 311	0,78	30 %	1,07	0,69
Lechwerke AG	Tyskland	16 694	0,01	30 %	0,28	0,28
Public Power Corporation SA	Hellas	14 351	0,59	26 %	1,09	0,76
Repower AG	Sveits	3 489	0,61	18 %	0,40	0,27
Societa Elettrica Sopracenerina SA SES	Sveits	1 033	0,53	18 %	0,13	0,09
SSE PLC	Storbritannia	139 940	0,82	23 %	0,42	0,26
VERBUND AG	Østerrike	21 448	0,68	25 %	0,74	0,49
Gjennomsnittsverdier			**0,72**			**0,40**
Median			**0,71**			**0,35**

Chapter 19: Valuation and Financial Modeling

Estimating the cost of capital

Market specific inputs	
Risk free rate	2,3 %
Risk premium	5,0 %
Expected inflation	2,5 %
Other inputs	
Tax level	27 %
Unlevered (asset) beta	0,4
Company specific inputs	
Size premium	3 %
Debt ratio [D/(E+D)]	35 %
Debt interest rate	4 %
Calculations	
Equity ratio	65 %
Debt to equity ratio	54 %
Levered (equity) beta	0,56

WACC	
re	8,1 %
Equity ratio	65 %
rd	4,0 %
Debt ratio	35 %
WACC	**6,3 %**

$$r_e = r_f + \beta_e \times (r_m - r_f) + \alpha = r_f + \beta_e \times RP + \alpha$$

$$WACC = r_e \times \frac{E}{D+E} + r_d \times \frac{D}{D+E} \times (1 - \tau)$$

Chapter 19: Valuation and Financial Modeling

Valuation – Discounted Cash Flow (DCF)

- Now that we have WACC we can discount the future cash flows to obtain the value of the company

		2010	2011	2012	2013	2014	2015	2016
Fri kontantstrøm					**2 245 988**	**2 358 288**	**2 476 202**	**2 600 012**
Neddiskonteringstid					0,50	1,50	2,50	3,50
WACC	6,3 %				0,97	0,91	0,86	0,81
Nåverdi av kontantstrømmer					2 178 857	2 153 081	2 127 611	2 102 442
Terminalverdi	2,5 %							35 275 998

- Discounting the free cash flow with WACC gives us the present value of the free cash flows
- At the end of our estimating period we use a terminal value consisting of a free cash flow in perpetuity. The estimated growth is equal to the inflation in this case

Chapter 19: Valuation and Financial Modeling

Valuation – Discounted Cash Flow (DCF)

		2010	2011	2012	2013	Prognose 2014	2015	2016
						Prognose		
		2010	**2011**	**2012**	**2013**	**2014**	**2015**	**2016**
Sum driftsinntekter		5 144 386	5 295 622	6 053 394	6 356 064	6 673 867	7 007 560	7 357 938
Driftskostnader								
Varekostnader		(1 401 087)	(1 432 914)	(1 845 065)	(1 937 318)	(2 034 184)	(2 135 893)	(2 242 688)
Personalkostnader		(601 150)	(694 114)	(781 720)	(820 806)	(861 846)	(904 939)	(950 186)
Andre driftskostnader		(1 017 785)	(157 054)	(732 780)	(769 419)	(807 890)	(848 284)	(890 699)
EBITDA		**2 124 364**	**3 011 540**	**2 693 829**	**2 828 520**	**2 969 946**	**3 118 444**	**3 274 366**
Avskrivninger		(675 110)	(780 769)	(841 113)	(883 169)	(927 327)	(973 693)	(1 022 378)
EBIT		**1 449 254**	**2 230 771**	**1 852 716**	**1 945 352**	**2 042 619**	**2 144 750**	**2 251 988**
Skatt		(405 791)	(624 616)	(518 760)	(544 699)	(571 933)	(600 530)	(630 557)
Driftsresultat etter skatt		**1 043 463**	**1 606 155**	**1 333 956**	**1 400 653**	**1 470 686**	**1 544 220**	**1 621 431**
Av- og nedskrivninger		675 110	780 769	841 113	883 169	927 327	973 693	1 022 378
Endring i arbeidskapital	12,5 %				(37 834)	(834 233)	(875 945)	(125 234)
Investeringer								
Fri kontantstrøm					**2 245 988**	**1 563 780**	**1 641 969**	**2 518 575**
Neddiskonteringstid					0,50	1,50	2,50	3,50
WACC	6,3 %				0,97	0,91	0,86	0,81
Nåverdi av kontantstrømmer					2 178 857	1 427 708	1 410 818	2 036 589
Terminalverdi	2,5 %							35 275 998

Chapter 19: Valuation and Financial Modeling

Valuation – Discounted Cash Flow (DCF)

- Adding the present value of the free cash flows and the terminal value of the cash flows in perpetuity gives us the total value of the company
- Note that cash and long term interest bearing debt is excluded from the calculations for simplicity

Verdiberegning	
+ Nåverdi prognoseperiode	8 561 991
+ Nåverdi terminalperiode	35 275 998
= Total selskapsverdi	**43 837 989**
+ Likvide midler og ikke driftsrelaterte aktiver	-
- Rentebærende gjeld	-
= Verdi av egenkapital	**43 837 989**

- The value of the company is hence estimated to be 43,8 mrd NOK

**The example is simplified for the purpose of this lecture*

Chapter 19: Valuation and Financial Modeling

Valuation – The Normalized Earnings Method

- An alternative, and a common valuation method for companies that have shown stable results over a period of time, is the normalized earnings method (normalresultatmetoden)
- The goal is to establish a normalized level of the company's result and then use this normalized level as the terminal for perpetuity
- A lot of in debt analysis is required to perform this analysis correctly
- With this method we have to adjust the WACC for inflation so it becomes real instead of nominal
- In the normalized earnings method, we use the inflation adjusted WACC (real WACC) as the discount factor in the perpetuity formula, with a growth equal to the inflation target of Norges Bank

Chapter 19: Valuation and Financial Modeling

Valuation – The Normalized Earnings Method

- For simplicity we assume that the normal result equals the result of 2012

	2009	2010	2011	2012	Normal-resultat
Sum driftsinntekter	4 466 660	5 248 232	5 336 233	6 053 394	6 053 394
Driftskostnader					
Varekostnader	(974 719)	(1 429 370)	(1 443 903)	(1 845 065)	(1 845 065)
Personalkostnader	(652 441)	(613 285)	(699 437)	(781 720)	(781 720)
Andre driftskostnader	(564 689)	(1 038 330)	(158 258)	(732 780)	(732 780)
EBITDA	**2 274 811**	**2 167 247**	**3 034 635**	**2 693 829**	**2 693 829**
Avskrivninger	(563 501)	(688 738)	(786 756)	(841 113)	(841 113)
EBIT	**1 711 310**	**1 478 509**	**2 247 878**	**1 852 716**	**1 852 716**

By applying the normalized earnings method we see that the value of the company is estaimted to be 36,4 mrd NOK

Verdiberegning	
Estimert normalisert driftsresultat	1 852 716
Skatt (28 %)	(518 760)
Estimert normalisert driftsresultat etter skatt	**1 333 956**
Beregnet verdi av normalresultat (WACC 3,7%)	**36 393 516**
+ Likvide midler og ikke driftsrelaterte aktiver	-
- Rentebærende gjeld	-
= Verdi av egenkapital	**36 393 516**

Chapter 19: Valuation and Financial Modeling

Valuation – The Normalized Earnings Method

External file

Chapter 19: Valuation and Financial Modeling

Sensitivity

- Valuations are sensitive to several of the parameters involved in the estimations, and it is common to calculate sensitivity analysis to see how the value change when key parameters change
- The WACC and the normalized result of the company are two parameters that often are analyzed with respect to sensitivity

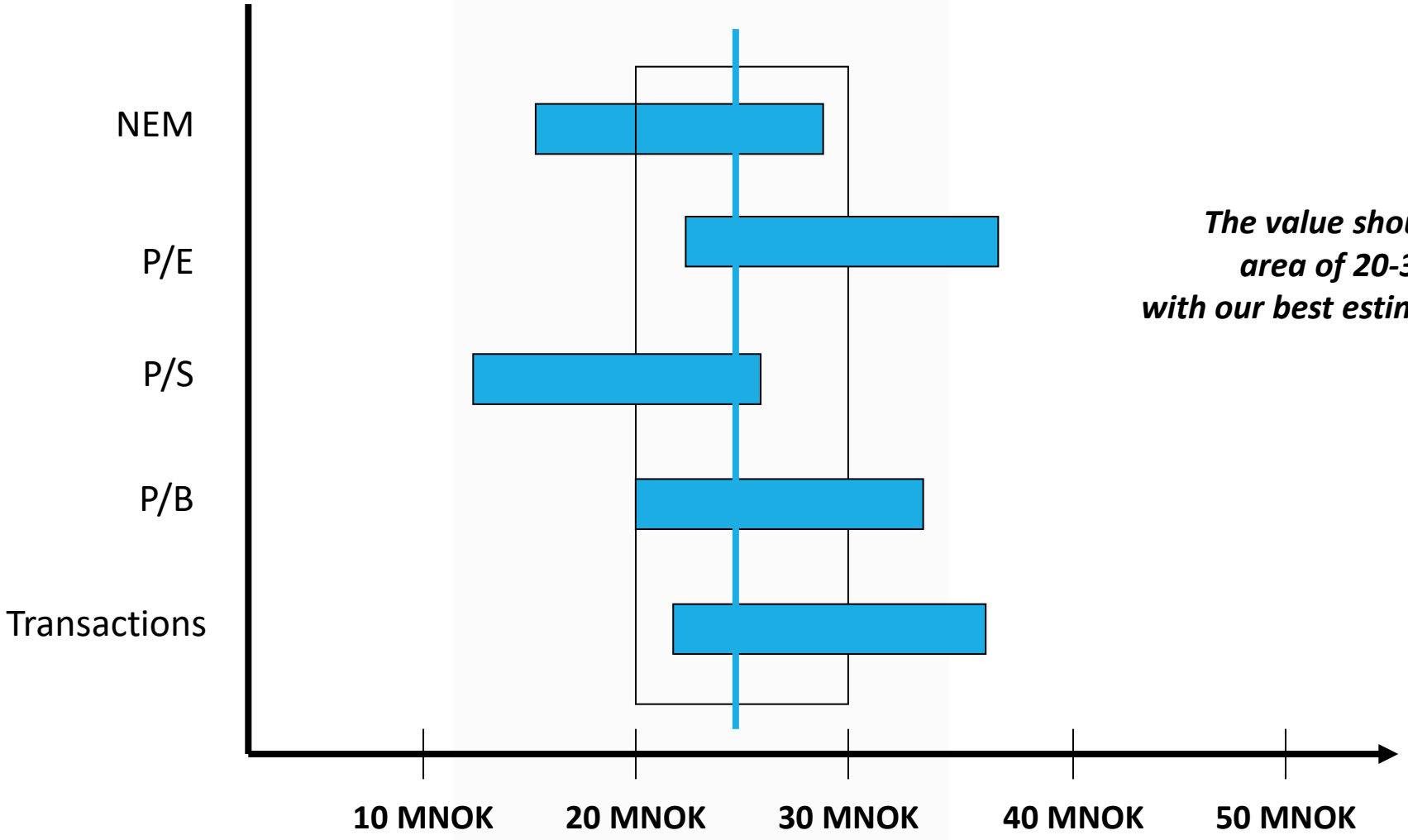
Sensitivitet normalresultat

Maks endring i WACC	1 %
Maks endring i normalresultat	20 %

		Normalresultat etter skatt				
		-20 %	**-10 %**	**Normal**	**10 %**	**20 %**
		1 067 164	**1 200 560**	**1 333 956**	**1 467 351**	**1 600 747**
WACC	**2,7 %**	40 038 196	45 042 971	50 047 746	55 052 520	60 057 295
	3,2 %	33 713 778	37 928 000	42 142 222	46 356 445	50 570 667
	3,7 %	29 114 813	32 754 165	**36 393 516**	40 032 868	43 672 220
	4,2 %	25 619 944	28 822 437	32 024 931	35 227 424	38 429 917
	4,7 %	22 874 185	25 733 458	28 592 731	31 452 004	34 311 277

Chapter 19: Valuation and Financial Modeling

Sensitivity



The value should be in the area of 20-30 MNOK, with our best estimate of 25 MNOK

Chapter 19: Valuation and Financial Modeling

Example from the housing market in Trondheim

How can you determine if the pricing of a house/apartment is fair?

EIE eiendomsmegling Trondheim

Dyre Halses gate 8, Trondheim

Solsiden - Fin 2-roms toppleilighet i særdeles ettertraktet område - God størrelse ...

47 m² 2 490 000 kr ??

Totalpris: 3 388 010 kr · Fellesutg.: 6 457 kr · Andel · Leilighet · 1 soverom

Visning - 22. mai, kl 18:45

Chapter 19: Valuation and Financial Modeling

Example from the housing market in Trondheim

How can you determine if the pricing of a house/apartment is fair?

- Call the broker and ask what the price should be:

«This is a fantastic opportunity for you!! Look at the brand new bathroom, amazing views ....blabla»

Chapter 19: Valuation and Financial Modeling

Example from the housing market in Trondheim

What about using some of the financial tools you have learned in this lecture?

Chapter 19: Valuation and Financial Modeling

Example from the housing market in Trondheim

How can you determine if the pricing of a house/apartment is fair?

- Estimate a yearly rent, calculate the implied yield and compare the implied yield to other investments.

Monthly rent: 15.000 NOK

Montly costs: 3.000 NOK

Free cash flow per month: 12.000 NOK

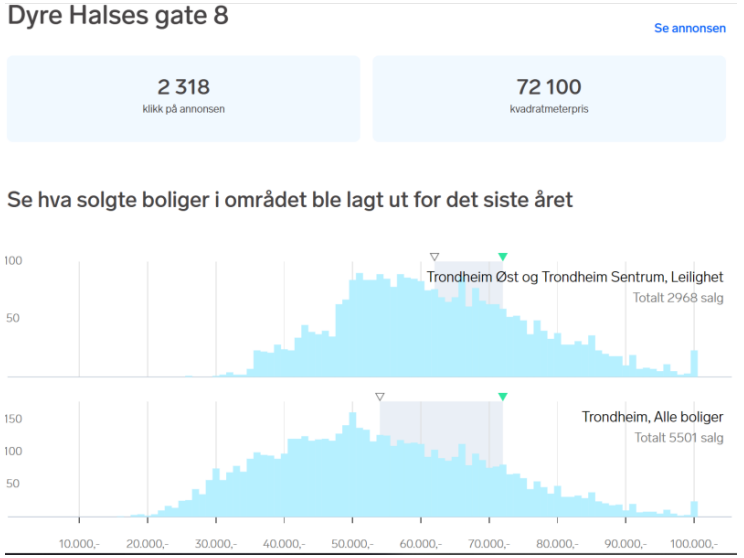
Yearly free cash flow: 144.000 NOK

- Look at the price per square meters of similar objects.

Calculate the estimatet price based on this multiple.

The apartment is 47 square meters

The average price per square meter for this area is 72 100 NOK



Chapter 19: Valuation and Financial Modeling

Example from the housing market in Trondheim

How can you determine if the pricing of a house/apartment is fair?

- Estimate a yearly rent, calculate the implied yield and compare the implied yield to other investments.

Monthly rent: 15.000 NOK

Montly costs: 3.000 NOK

Free cash flow per month: 12.000 NOK

Yearly free cash flow: 144.000 NOK

The implied yield given the listing price would be 4,25 %.

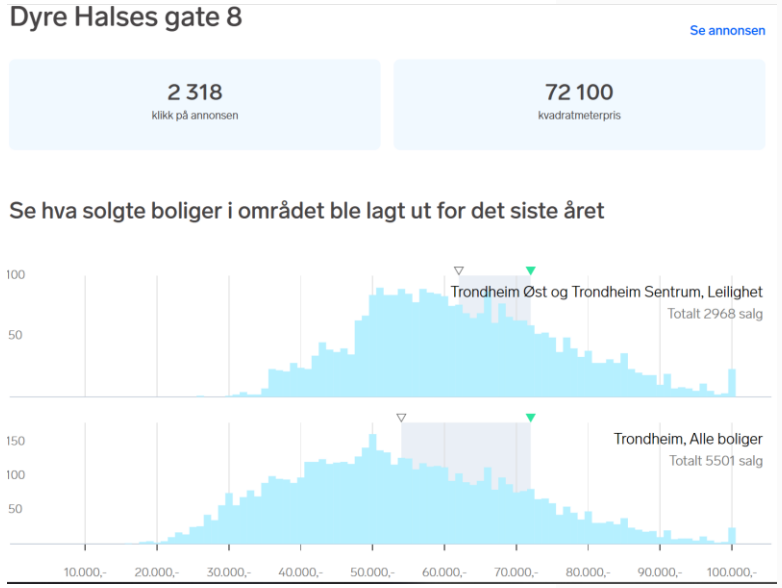
What does this tell us about the current housing market?

Chapter 19: Valuation and Financial Modeling

Example from the housing market in Trondheim

How can you determine if the pricing of a house/apartment is fair?

- Look at the price per square meters of similar objects. This is basically the same as the multiples we considered in the Ideko example.



- The apartment is 47 square meters
- The average price per square meter for this area is 72 100 NOK
- By applying this multiple we find that the value of the apartment based on similar prospects would be:

47 sqm x 72 100 NOK/sqm= 3.388.700 NOK

Case study: Company Valuation

Case study: Company Valuation

- Typical assignments for a financial advisor
  - Financing (projects, equity and debt)
  - Valuations
  - Financial analysis and model construction
  - M&A
  - Management buyout
  - Legal advisors

- Cost of capital
  - Capital Asset Pricing Model: $r_e = r_f + \beta(r_m - r_f) = r_f + \beta \cdot RP$ where $\beta = \frac{\rho_{R,Rm} \cdot \sigma_R}{\sigma_{Rm}}$
  - Parametere to decide: r_f, r_m, r_g and β
  - Weighted Average Cost of Capital: $WACC = [E/(E+G)] \cdot r_e + [G/(E+G)] \cdot r_g \cdot (1-\tau)$

- Valuations
  - Buyer, seller or legal adviser?
  - Definitely not an exact science!

Case study: Company Valuation

- Valuations are performed to get the correct value of a business
  - «As-is» → How the business is today
  - «To-be» → Restructuring and changes in the company

- «As-is» valuations are often related to
  - Mergers and acquisitions
  - Change in value between two periods in time
  - Measure of efficiency improvement
  - Impairment tests

- «To-be» valuations are often related to
  - Strategic change in the company
  - Change in the market situation
  - Investment decisions

- Most of the valuations are a combination of «as-is» and «to-be»
  - Status quo in combination with an estimate on present value of growth opportunities

Case study: Company Valuation

«Will you need any of this stuff when you start working after finishing at NTNU?»

CAPM

Discounted cashflow (DCF)

Required return

Multiples
(P/E, EV/EBIT osv)

Flow of dividends

Normalized earnings method

WACC

Case study: Company Valuation

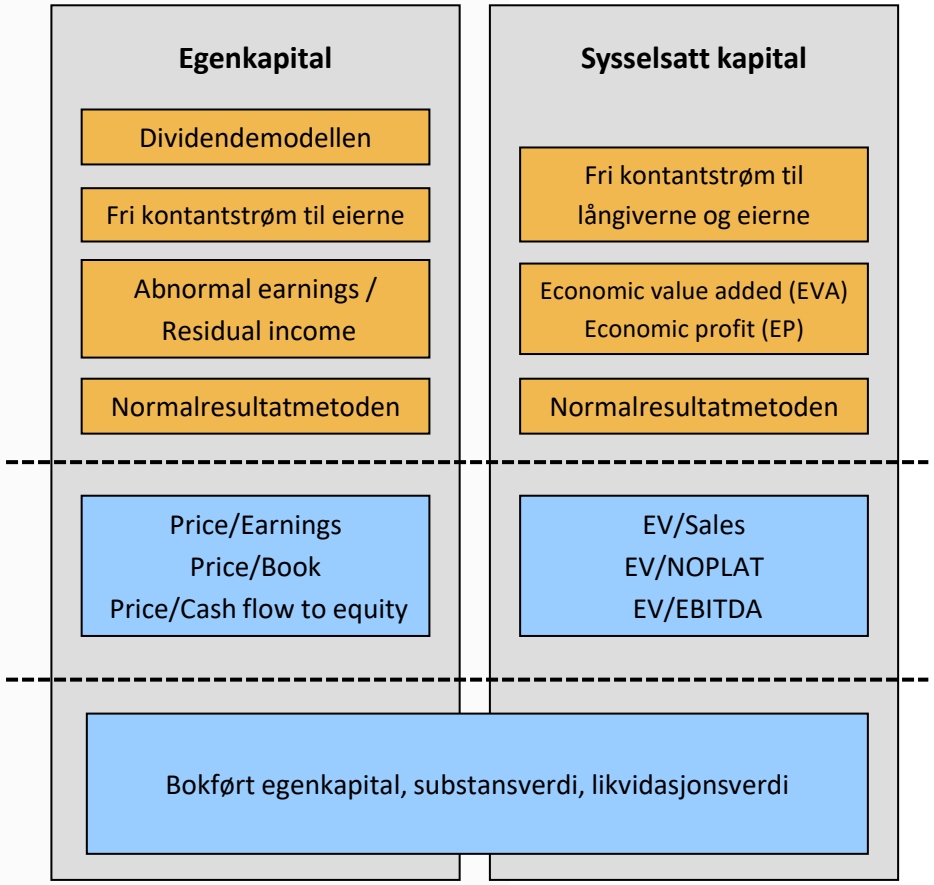
- **Inntjeningsbaserte tilnærminger** er basert på neddiskontering av forventede fremtidige kontantstrømmer

Fordeler

- I det lange løp er ikke en virksomhet verdt mer enn den verdien den kan forrente
- Verdien er avhengig av investorens avkastningskrav
- Bedriftsøkonomisk verdi er den verdien virksomheten kan forrente for eierne diskontert med valgt avkastningskrav

Ulemper

- Fremtidig inntjening er alltid usikker
- Fastsettelse av avkastningskrav blir ofte tilfeldig
- Det er vanskelig å vite hvor lenge inntjeningen kan opprettholdes og hvilke investeringer som må gjøres for å opprettholde inntjeningen



Case study: Company Valuation

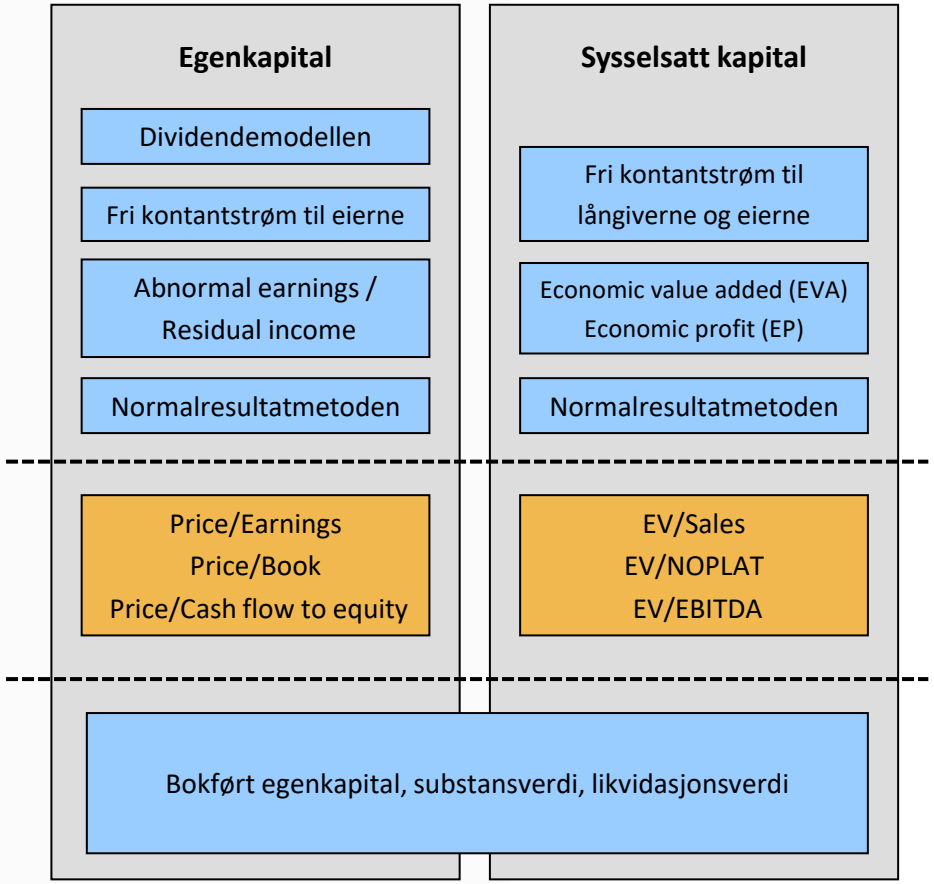
- **Markedsorienterte tilnærminger** er basert på sammenligning med verdien av lignende selskaper

Fordeler

- Data og sammenligningsgrunnlag er lett tilgjengelig gjennom børser og offentlige arkiver
- Kan være enklere å selge inn til en beslutningstaker da tilnærmingen er enkel og forståelig

Ulemper

- Ulik egnethet fra bransje til bransje
- Underliggende prosesser og aktiviteter tas ikke hensyn til
- Forutsetter at markedene handler på perfekt informasjon, noe som sjelden er tilfelle



Case study: Company Valuation

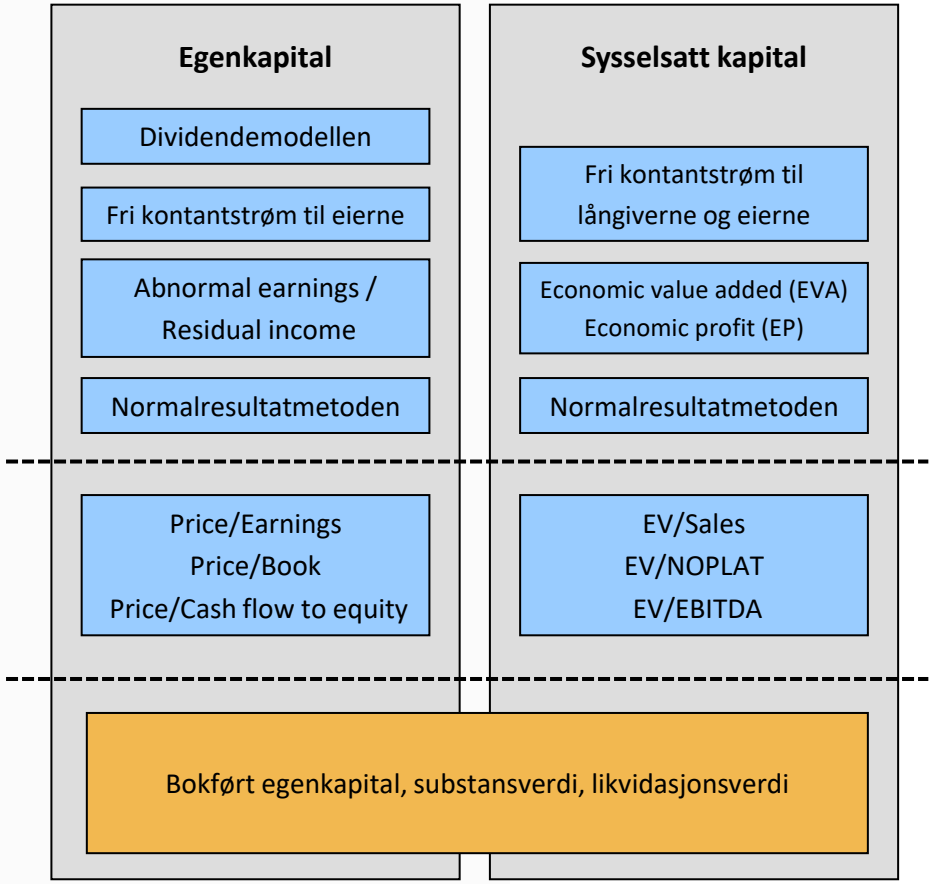
- **Balansetilnærminger** viser eiendelenes verdi i et annenhåndsmarket

Fordeler

- Hvis det finnes et annenhåndsmarked så har man en faktisk verdi å forholde seg til
- Underliggende eiendeler er ofte lettere å realisere enn forventninger om fremtidig inntjening

Ulemper

- Verdidrivere i en virksomhet består av langt mer enn underliggende eiendeler. Kompetanse, organisasjon, nettverk og teknologi prises ikke inn i balansebaserte metoder



Case study: Company Valuation

External file

Case study: Information Memorandum

External file

Introduction to the mandatory exercise

Mandatory exercise: TIØ 4145 Corporate Finance

VALUATION OF RADIONOR COMMUNICATIONS AS

Due Monday 27th of October 2025

To be delivered in groups
of maximum 2 students

Mandatory Exercise

Valuation of Radionor Communications AS

Perform a valuation of Radionor Communications AS. The valuation should be delivered as a report (max 10 pages) including necessary calculations, financial modeling and explanations of the assumptions made.

Objectives:

1) Make a one-pager company profile, including a short description of the business and the market.

2) Present a short summary the *various valuation methods* that can be used for this valuation.

3) Find the Weighted Average Cost of Capital (WACC) for the company.
   - Calculate/estimate the following parameters. Show the calculations and explain your assumptions.
     - Beta (β)
     - Risk free rate (r_f) and market returns (r_m)
     - Equity cost of capital (r_e)
     - Cost of debt (r_d)

4) Estimate an interval for the market capitalization of the company by applying multiples of comparable companies world-wide.

Mandatory Exercise

Valuation of Radionor Communications AS

5) Value the company by applying the discounted free cash flow model (DCF).
   - Determine a 5 year budget/forecast for the company and establish a financial model based on the forecast
   - Find the free cash flows and the terminal value
   - Calculate the present value of the free cash flows and estimate the market capitalization

6) Value the company by applying the normalized earnings method (NEM).
   - Establish a normalized EBIT for the company (see attachment for procedure)
   - Calculate the present value of the free cash flow and estimate the market capitalization

7) Find the implied normalized EBITDA that reflects a share price of NOK 20 billion for the company. Explain your assumptions needed to obtain the normalized EBITDA.

8) Perform a sensitivity analysis with respect to changes in WACC and EBITDA.

9) List the Equity values that you have obtained from your various calculations. Discuss the various values and conclude with your recommendation of what the correct value of the company is.

10) Make a 1 page “Investor teaser” of the company that can be presented to potential investors if the company were to be sold

Thank you!